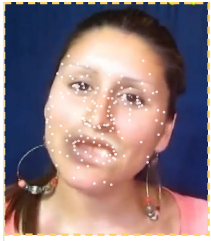
1 Photo, face crop

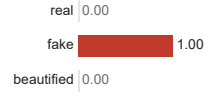


478 MediaPipe landmarks
(one in three drawn)
aligned crop, as in training
runs in the browser

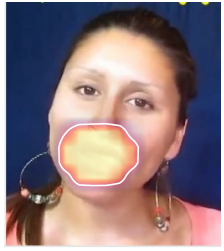
2 Detector and evidence head

CLIP ViT-L/14 + LoRA

r16: 3.2 M of 306 M trained
3 class scores, 16×16 evidence
smoothing score (if beautified)

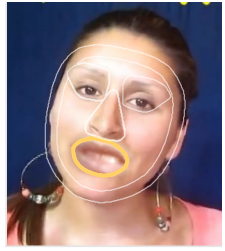


class scores (softmax)
decision = the largest



evidence head: 16×16 map,
upsampled; bright = strong
evidence, white line = 0.5

3 Part naming (code)



five part masks from landmarks
highest mean evidence is named:
mouth 0.98 (nose 0.42, skin 0.27,
eyes 0.01, contour 0.04)

4 Sentences, checks (code)

record

```
decision   fake
p          0.00 / 1.00 / 0.00
evidence   eyes 0.01  nose 0.42
           mouth 0.98  skin 0.27
named      mouth
```

sentences from templates

- ✓ The model's decision is fake (score 1.00).
- ✓ The evidence is concentrated on the mouth (mean evidence 0.98).
- ✓ Scores are raw model outputs, not calibrated probabilities.

checks H1–H6: score vs threshold,
all facts present, no certainty claim,
no score attached to a part

validation: mouth restored from the
original, forgery score 1.00 → 0.08

neural network code (fixed rules) checks

example: FF++ test frame 454 (genuine) with the mouth replaced from a donor; part-level forgery test set, not seen in training.